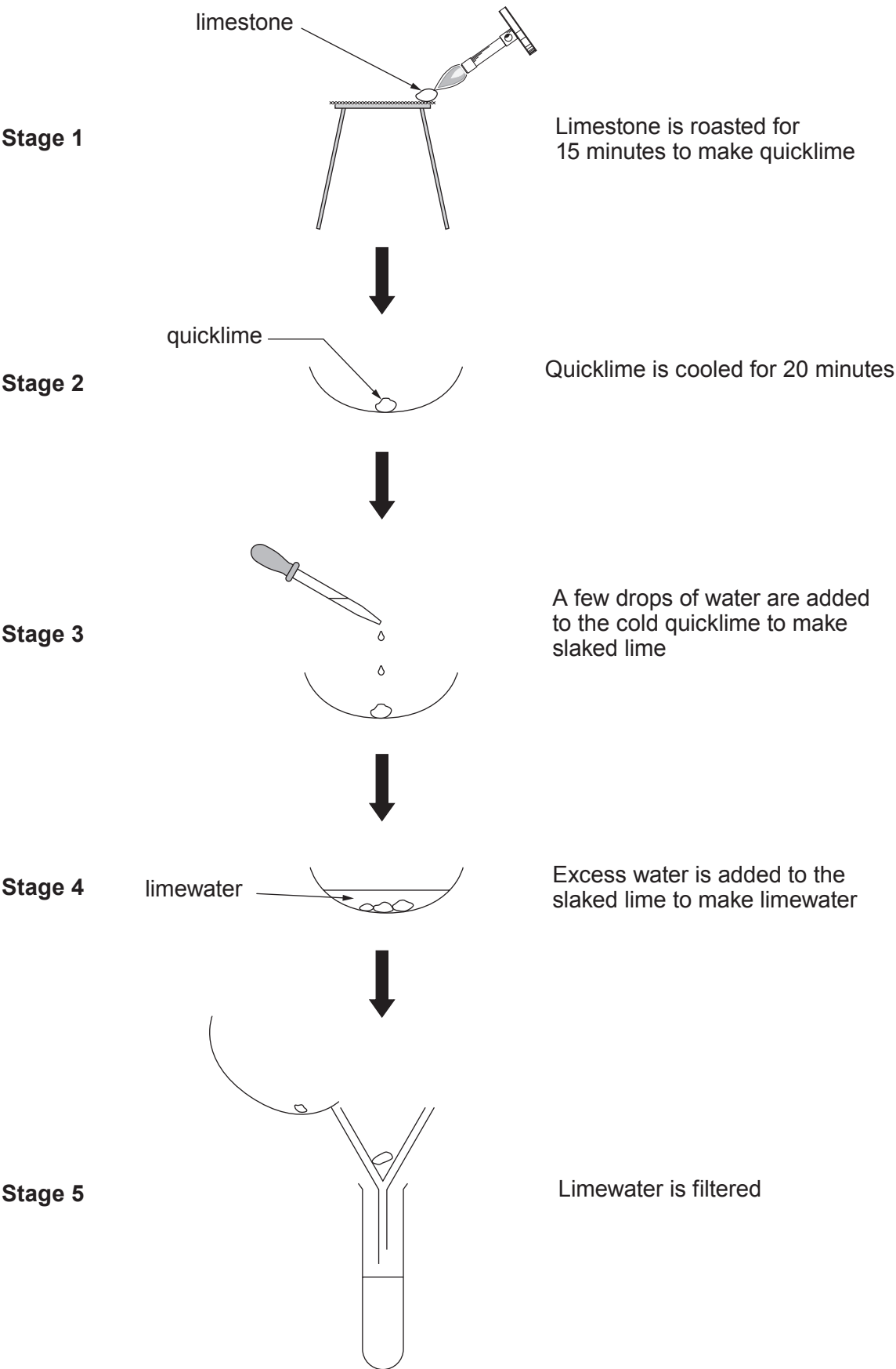


7. The flow chart shows the stages a student carried out to change calcium carbonate (limestone) into calcium hydroxide solution (limewater).



(a) Use the information in the flow chart to answer parts (i) to (iii).

(i) Name the type of reaction taking place in stage 1.

[1]

.....

(ii) Give the **number** of the stage which shows an exothermic reaction.

[1]

.....

(iii) I. Describe what you would expect to **see** happen in stage 3.

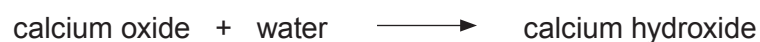
[2]

.....

.....

.....

II. The reaction taking place in stage 3 is shown by the following word equation.

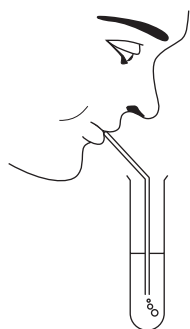


Write a balanced **symbol** equation for this reaction.

[2]

..... + $\longrightarrow$

(b) The student blew gently through a straw placed in a test tube containing limewater.



Describe what you would expect to happen to the limewater. Give a reason for your answer.

[2]

Observation

Reason

- (c) Quicklime is manufactured from limestone in a lime kiln. A manufacturer expected to obtain 5.6 tonnes per 10 tonnes of limestone used. The actual mass of quicklime produced was 5.1 tonnes.

Calculate the percentage yield of quicklime for this manufacturing process. [2]

percentage yield = %

Examiner
only

10

7. The following information is taken from some articles about global warming.

Greenhouse gases such as carbon dioxide keep heat close to the Earth’s surface making it a suitable temperature for life. Global warming is an increase of the Earth’s mean surface temperature due to the overproduction of greenhouse gases by burning fossil fuels such as gas, petrol and oil. Deforestation also contributes to this. With the growth of industry in the 1900s, humans began burning more fossil fuels to run our cars, trucks and factories. There is more carbon dioxide in the atmosphere today than at any point in the last 800,000 years.

The following charts show the mean global temperature every decade since the 1880s, the amount of carbon dioxide in the atmosphere from 1750-2010 and the main sources of carbon dioxide production today.

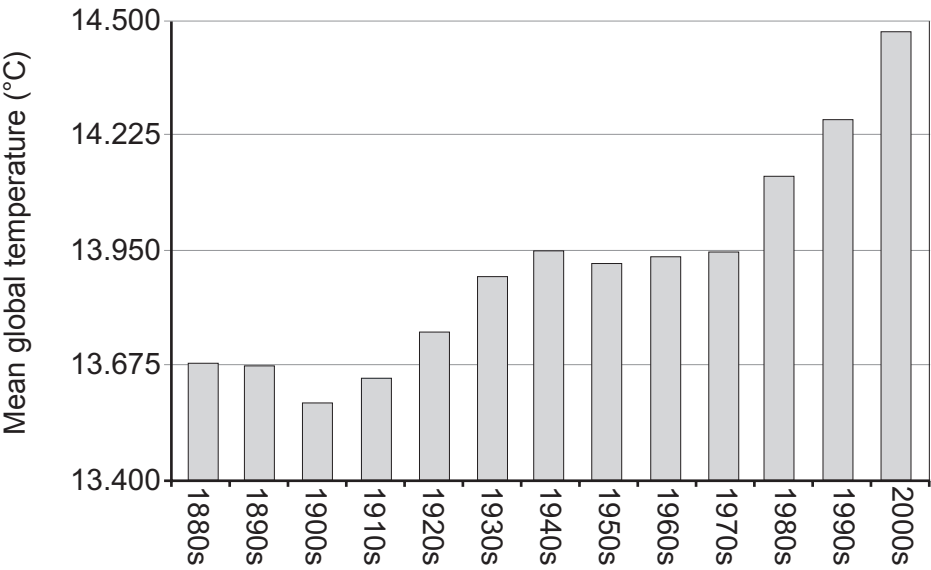


Figure 1

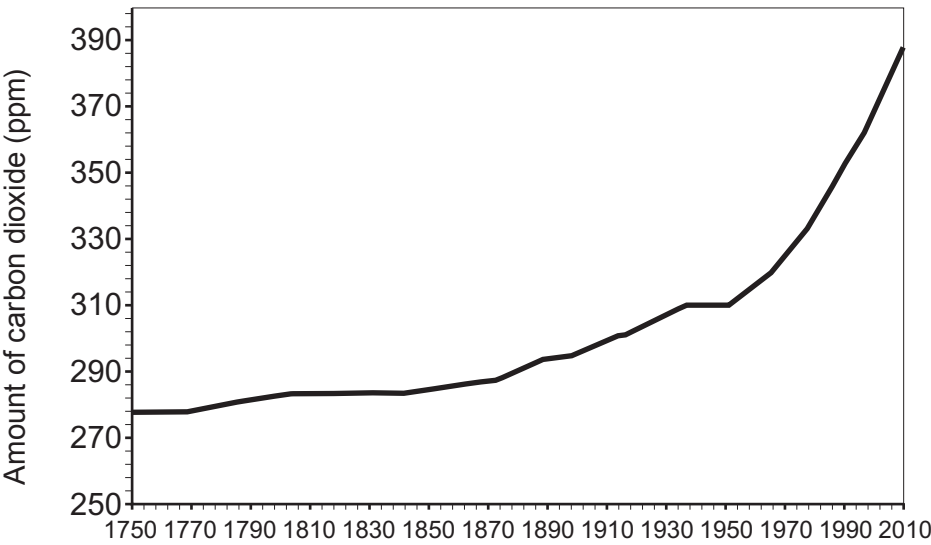


Figure 2



Examiner
only

7. (a) The water used in our homes is treated to make it safe to drink.

State the main stages involved in the treatment of the public water supply. Describe the purpose of each stage. [6 QER]

You do not need to refer to fluoridation in your answer.

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Examiner
only

(b) Fluoride is sometimes added to water supplies.

State why fluoride is added and give **one** reason why some people oppose the addition of fluoride to water. [2]

Reason for adding fluoride

.....

.....

Reason why some people oppose its addition

.....

.....

(c) This advert was published to encourage people to save water by taking a shower instead of a bath.

To
Shower
or to
Bathe?

A shower for
8 minutes
uses 76 litres



A full bath
uses 160 litres

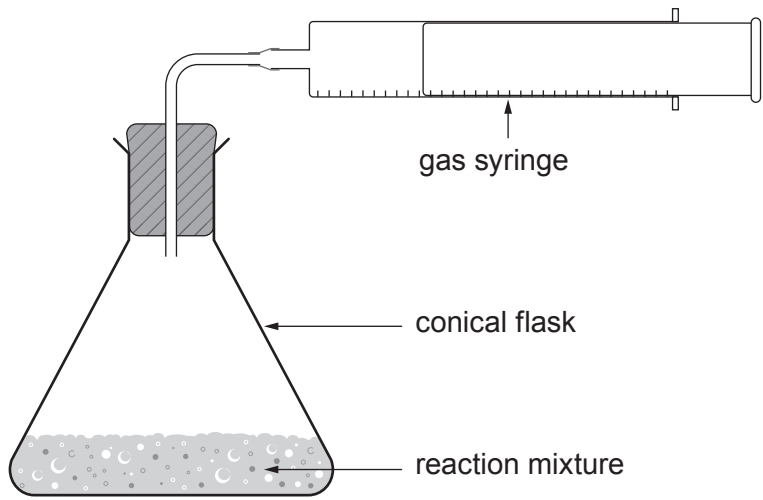
Use the data in the advert to calculate the percentage of water **saved** by taking a shower for 8 minutes rather than running a full bath. [2]

Percentage of water saved = %

10



7. A student decided to investigate the rate of reaction of different concentrations of hydrochloric acid with pieces of chalk, using the equipment shown.



Chalk contains calcium carbonate and the reaction produces carbon dioxide gas.
The table shows the student's results.

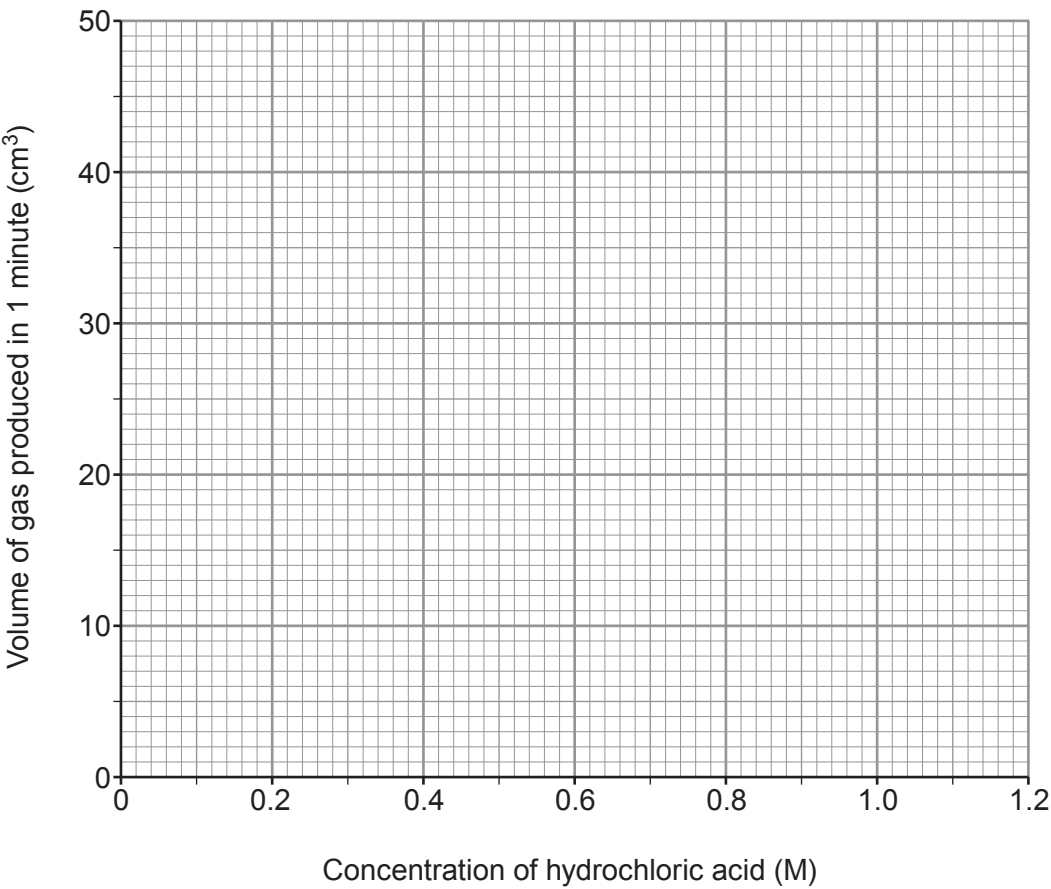
Concentration of hydrochloric acid (M)	Volume of gas produced in 1 minute (cm ³)
0.2	8
0.4	16
0.6	24
0.8	32
1.0	40
1.2	48



(a) Plot the results from the table on the grid. Draw a suitable line.

[3]

Examiner
only



Examiner
only

- (b) Describe how the volume of gas produced in 1 minute changes as the concentration of the acid changes. [2]

.....

.....

.....

- (c) Underline the correct word(s) in each bracket to explain the results. [3]

When the acid concentration is higher, there are
(**more** / less / the same number of) particles in the same volume.

Acid particles (**dissolve** / mix / collide) with the chalk more frequently.

The (**chalk** / acid / gas) is produced at a higher rate.

- (d) Other than changing the concentration of the acid or the mass of the chalk added, state **two** ways in which the rate of this reaction could be increased. [2]

.....

.....

.....

10



7. Is it right to waste helium on party balloons?



Helium is a colourless inert gas found in Group 0 of the Periodic Table.

Helium is one of the commonest elements in the Universe, second only to hydrogen. However, on Earth it is relatively rare, as shown in **Table 1**.

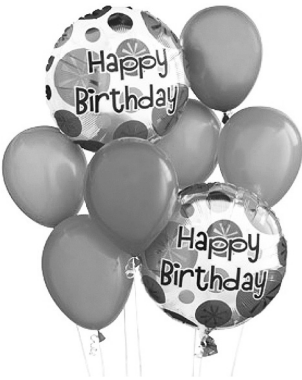
Gases which have a density less than air can escape the Earth’s gravity and leak away into space. The density of air is 1.2g/m³. **Bar chart 1** shows the densities of Group 0 gases.

Helium has the lowest boiling point of any element. This makes it of key importance for magnets used in hospital MRI scanners, which must be super-cooled to generate the hugely powerful magnetic fields required.

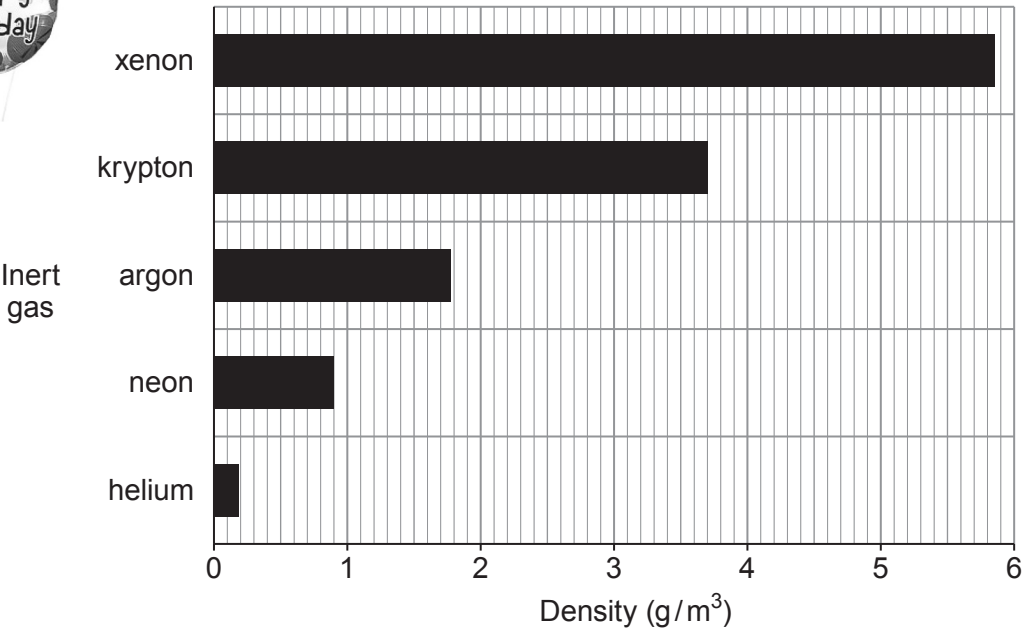
Some scientists believe that because helium is a finite resource it should not be used for party balloons.

Table 1

Inert gas	Percentage in the atmosphere (%)	Melting point (°C)	Boiling point (°C)
helium	0.00052	-272	-269
neon	0.0018	-246	-246
argon	0.93	-186	-186
krypton	0.0001	-152	-152
xenon	0.000009	-111	-106



Bar chart 1



(a) Answer the following questions using the information given.

- (i) Tick (✓) the box next to the **most** important property that makes helium a suitable material to fill **floating** party balloons. [1]

Helium is a gas

☐

Helium is the second most common element in the Universe

☐

Helium is less dense than air

☐

Helium is colourless

☐

- (ii) Tick (✓) the box next to the correct statement. [1]

The Earth's atmosphere contains more helium than argon

☐

The Earth's atmosphere contains more xenon than helium

☐

The Earth's atmosphere contains more helium than krypton

☐

- (iii) Tick (✓) the box next to the **best** reason for not using helium to fill party balloons. [1]

There isn't much helium in the Earth's atmosphere

☐

Scientists say helium shouldn't be used to fill balloons

☐

Helium is a finite resource

☐

- (iv) Tick (✓) the box next to the correct statement. [1]

Only helium gas can leak away into space

☐

Helium and neon gases can leak away into space

☐

Only argon can leak away into space

☐

All inert gases can leak away into space

☐

(b) The table below shows the electronic structure of three Group 0 elements.

Group 0 element	Electronic structure
helium	2
neon	2,8
argon	2,8,8

Tick (✓) the box next to the statement that **best** explains why Group 0 elements are unreactive.

[1]

All Group 0 elements have 2 electrons in their inner shell

☐

All Group 0 elements have 8 electrons in their outer shell

☐

All Group 0 elements have full outer shells

☐

All Group 0 elements have some full shells

☐

5

